ONLINE SHOPPING SYSTEM

UML Class Diagram Report

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UML Class Diagram Report

1. Introduction

This report describes the UML Class Diagram for the proposed Online Shopping System. The diagram shows the system's classes, attributes, operations, relationships, and multiplicities.

2. Classes, Attributes, and Methods

Customer

Attributes: customerId: int, name: String, email: String, address: String

Methods: register(): void, updateProfile(): void

Product

Attributes: productId: int, name: String, price: double, stockQuantity: int

Methods: displayDetails(): void, updateStock(quantity: int): void

ShoppingCart

Attributes: cartId: int, totalAmount: double

Methods: addItem(product: Product, quantity: int): void, removeItem(item: CartItem): void, calculateTotal(): double

CartItem

Attributes: quantity: int, unitPrice: double

Methods: calculateSubtotal(): double

Order

Attributes: orderId: int, orderDate: Date, status: String, totalAmount: double

Methods: placeOrder(): void, cancelOrder(): void, calculateTotal(): double

OrderItem

Attributes: quantity: int, unitPrice: double

Methods: calculateSubtotal(): double

Payment

Attributes: paymentId: int, amount: double, method: String, paymentDate: Date, status: String

Methods: processPayment(): boolean, refundPayment(): boolean

3. Relationships and Multiplicities

1. Customer — ShoppingCart (1 to 1): each customer has one shopping cart; each cart belongs to one customer.

2. Customer — Order (1 to 0..*): a customer may place zero or many orders; each order belongs to one customer.

3. ShoppingCart — CartItem (1 to 0..*): a cart may contain zero or many cart items; each cart item belongs to one cart. This is modelled as composition because cart items are part of a cart.

4. Product — CartItem (1 to 0..*): each cart item refers to one product; a product may appear in many cart items.

5. Order — OrderItem (1 to 1..*): an order contains one or more order items; each order item belongs to one order. This is modelled as composition because order items are part of an order.

6. Product — OrderItem (1 to 0..*): each order item refers to one product; a product may appear in many order items.

7. Order — Payment (1 to 0..1): an order may have no payment record yet or one payment record; each payment record is associated with one order.

4. Design Assumptions

• Each customer has exactly one shopping cart in this model.

• Each order contains at least one order item.

• Each order has at most one payment record. The model does not cover split payments or multiple payment attempts.

• A cart item and an order item each refer to one product.

• The diagram is a conceptual design and does not implement actual payment processing.

5. Conclusion

The class model separates customer, product, cart, order, and payment responsibilities. Multiplicities clarify how many objects may participate in each relationship.